



# Dimensional formulae

Basic quantities, electrical principles, fluid dynamics and heat transfer

**BASE DIMENSIONS**

M = mass    L = length    T = time  
I = electric current     $\Theta$  = temperature

| Physical quantity                    | Symbol         | Relationship                                                             | Dimensions                                               | SI unit                            |
|--------------------------------------|----------------|--------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------|
| Length / displacement                | L, x, s        | Base quantity                                                            | [L]                                                      | m                                  |
| Mass                                 | m              | Base quantity                                                            | [M]                                                      | kg                                 |
| Time                                 | t              | Base quantity                                                            | [T]                                                      | s                                  |
| Electric current                     | I              | Base quantity                                                            | [I]                                                      | A                                  |
| Temperature                          | T              | Base quantity                                                            | [ $\Theta$ ]                                             | K                                  |
| Characteristic diameter / length     | D              | Length scale of the system                                               | [L]                                                      | m                                  |
| Area                                 | A              | Length $\times$ length                                                   | [L <sup>2</sup> ]                                        | m <sup>2</sup>                     |
| Volume                               | V              | Length <sup>3</sup>                                                      | [L <sup>3</sup> ]                                        | m <sup>3</sup>                     |
| Velocity / fluid velocity            | v, u           | Displacement / distance $\div$ time                                      | [LT <sup>-1</sup> ]                                      | m s <sup>-1</sup>                  |
| Acceleration                         | a              | Change in velocity $\div$ time                                           | [LT <sup>-2</sup> ]                                      | m s <sup>-2</sup>                  |
| Force                                | F              | Mass $\times$ acceleration                                               | [MLT <sup>-2</sup> ]                                     | N                                  |
| Density / fluid density              | $\rho$         | Mass $\div$ volume                                                       | [ML <sup>-3</sup> ]                                      | kg m <sup>-3</sup>                 |
| Pressure                             | p              | Normal force $\div$ area                                                 | [ML <sup>-1</sup> T <sup>-2</sup> ]                      | Pa                                 |
| Work / energy                        | W, E           | Force $\times$ distance (along force)                                    | [ML <sup>2</sup> T <sup>-2</sup> ]                       | J                                  |
| Power                                | P              | Energy transferred $\div$ time                                           | [ML <sup>2</sup> T <sup>-3</sup> ]                       | W                                  |
| Momentum                             | p              | Mass $\times$ velocity                                                   | [MLT <sup>-1</sup> ]                                     | kg m s <sup>-1</sup>               |
| Frequency                            | f              | 1 $\div$ period                                                          | [T <sup>-1</sup> ]                                       | Hz                                 |
| Charge                               | Q              | Current $\times$ time                                                    | [IT]                                                     | C                                  |
| Voltage                              | V              | Energy transferred $\div$ charge                                         | [ML <sup>2</sup> T <sup>-3</sup> I <sup>-1</sup> ]       | V                                  |
| Resistance                           | R              | Voltage $\div$ current                                                   | [ML <sup>2</sup> T <sup>-3</sup> I <sup>-2</sup> ]       | $\Omega$                           |
| Volumetric flow rate                 | Q              | Volume $\div$ time                                                       | [L <sup>3</sup> T <sup>-1</sup> ]                        | m <sup>3</sup> s <sup>-1</sup>     |
| Mass flow rate                       | $\dot{m}$      | Mass $\div$ time                                                         | [MT <sup>-1</sup> ]                                      | kg s <sup>-1</sup>                 |
| Dynamic viscosity                    | $\mu$          | Shear stress $\div$ velocity gradient                                    | [ML <sup>-1</sup> T <sup>-1</sup> ]                      | Pa s                               |
| Kinematic viscosity                  | $\nu$          | Dynamic viscosity $\div$ density                                         | [L <sup>2</sup> T <sup>-1</sup> ]                        | m <sup>2</sup> s <sup>-1</sup>     |
| Pressure gradient                    | dp/dx          | Change in pressure $\div$ distance                                       | [ML <sup>-2</sup> T <sup>-2</sup> ]                      | Pa m <sup>-1</sup>                 |
| Reynolds number                      | Re             | $\rho v L \div \mu$                                                      | [1]                                                      | Dimensionless                      |
| Convective heat-transfer coefficient | h              | Heat-transfer rate $\div$ (area $\times$ temperature difference)         | [MT <sup>-3</sup> $\Theta$ <sup>-1</sup> ]               | W m <sup>-2</sup> K <sup>-1</sup>  |
| Thermal conductivity                 | k              | Heat flux $\div$ temperature gradient                                    | [MLT <sup>-3</sup> $\Theta$ <sup>-1</sup> ]              | W m <sup>-1</sup> K <sup>-1</sup>  |
| Specific heat capacity               | c <sub>p</sub> | Heat added $\div$ (mass $\times$ temperature rise), at constant pressure | [L <sup>2</sup> T <sup>-2</sup> $\Theta$ <sup>-1</sup> ] | J kg <sup>-1</sup> K <sup>-1</sup> |

Reading the table: [LT<sup>-1</sup>] means length  $\div$  time. [1] means dimensionless.

Symbols: p, Q, V and T have different meanings in different contexts; check the quantity.